5737 Assignment 1 Question 1

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2025-02-12

Begin by installing and loading the necessary packages:

```r
# install.packages("stargazer")
# install.packages("dplyr")
# install.packages("glmnet")
# install.packages("e1071")
# install.packages("caret")
# install.packages("infotheo")

library(stargazer)
```

```
##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(glmnet)
```

```
## Loading required package: Matrix

## Loaded glmnet 4.1-8
```

```r
library(e1071)
library(caret)
```

```
## Loading required package: ggplot2

## Loading required package: lattice
```

```r
library(infotheo)
```

Load the dataset Happiness2015.csv. Note that you will need to choose it yourself.

```r
data <- read.csv(file.choose(), sep = ',')
```

Print a subset of the rows of the data. We'll observe just the first ten rows.

```r
head(data, n = 10)
```

```
##        Country                    Region Happiness.Rank Happiness.Score
## 1  Switzerland            Western Europe              1           7.587
## 2      Iceland            Western Europe              2           7.561
## 3      Denmark            Western Europe              3           7.527
## 4       Norway            Western Europe              4           7.522
## 5       Canada             North America              5           7.427
## 6      Finland            Western Europe              6           7.406
## 7  Netherlands            Western Europe              7           7.378
## 8       Sweden            Western Europe              8           7.364
## 9  New Zealand Australia and New Zealand              9           7.286
## 10   Australia Australia and New Zealand             10           7.284
##    Standard.Error Economy..GDP.per.Capita.  Family Health..Life.Expectancy.
## 1         0.03411                  1.39651 1.34951                  0.94143
## 2         0.04884                  1.30232 1.40223                  0.94784
## 3         0.03328                  1.32548 1.36058                  0.87464
## 4         0.03880                  1.45900 1.33095                  0.88521
## 5         0.03553                  1.32629 1.32261                  0.90563
## 6         0.03140                  1.29025 1.31826                  0.88911
## 7         0.02799                  1.32944 1.28017                  0.89284
## 8         0.03157                  1.33171 1.28907                  0.91087
## 9         0.03371                  1.25018 1.31967                  0.90837
## 10        0.04083                  1.33358 1.30923                  0.93156
##    Freedom Trust..Government.Corruption. Generosity Dystopia.Residual
## 1  0.66557                       0.41978    0.29678           2.51738
## 2  0.62877                       0.14145    0.43630           2.70201
## 3  0.64938                       0.48357    0.34139           2.49204
## 4  0.66973                       0.36503    0.34699           2.46531
## 5  0.63297                       0.32957    0.45811           2.45176
## 6  0.64169                       0.41372    0.23351           2.61955
## 7  0.61576                       0.31814    0.47610           2.46570
## 8  0.65980                       0.43844    0.36262           2.37119
## 9  0.63938                       0.42922    0.47501           2.26425
## 10 0.65124                       0.35637    0.43562           2.26646
```

Noting the column names of some variables, let's rename them for simplicity.

```r
names(data)[names(data) == "Happiness.Rank"] <- "rank"
names(data)[names(data) == "Happiness.Score"] <- "score"
names(data)[names(data) == "Standard.Error"] <- "standard_error"
names(data)[names(data) == "Economy..GDP.per.Capita."] <- "GDP"
names(data)[names(data) == "Family"] <- "family"
names(data)[names(data) == "Health..Life.Expectancy."] <- "health"
names(data)[names(data) == "Freedom"] <- "freedom"
names(data)[names(data) == "Trust..Government.Corruption."] <- "gov_trust"
names(data)[names(data) == "Generosity"] <- "generosity"
names(data)[names(data) == "Dystopia.Residual"] <- "dystopia_residual"
```

Produce a new dataset from the original, where the first 2 columns are no longer present.

```r
new_data <- data[-c(1,2)]
```

Now produce a summary of the data.

```r
summary(new_data)
```

```
##       rank            score       standard_error         GDP
##  Min.   :  1.00   Min.   :2.839   Min.   :0.01848   Min.   :0.0000
##  1st Qu.: 40.25   1st Qu.:4.526   1st Qu.:0.03727   1st Qu.:0.5458
##  Median : 79.50   Median :5.232   Median :0.04394   Median :0.9102
##  Mean   : 79.49   Mean   :5.376   Mean   :0.04788   Mean   :0.8461
##  3rd Qu.:118.75   3rd Qu.:6.244   3rd Qu.:0.05230   3rd Qu.:1.1584
##  Max.   :158.00   Max.   :7.587   Max.   :0.13693   Max.   :1.6904
##      family           health          freedom         gov_trust
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.00000
##  1st Qu.:0.8568   1st Qu.:0.4392   1st Qu.:0.3283   1st Qu.:0.06168
##  Median :1.0295   Median :0.6967   Median :0.4355   Median :0.10722
##  Mean   :0.9910   Mean   :0.6303   Mean   :0.4286   Mean   :0.14342
##  3rd Qu.:1.2144   3rd Qu.:0.8110   3rd Qu.:0.5491   3rd Qu.:0.18025
##  Max.   :1.4022   Max.   :1.0252   Max.   :0.6697   Max.   :0.55191
##    generosity     dystopia_residual
##  Min.   :0.0000   Min.   :0.3286
##  1st Qu.:0.1506   1st Qu.:1.7594
##  Median :0.2161   Median :2.0954
##  Mean   :0.2373   Mean   :2.0990
##  3rd Qu.:0.3099   3rd Qu.:2.4624
##  Max.   :0.7959   Max.   :3.6021
```

```r
stargazer(new_data, type = "text")
```

```
##
## ==================================================
## Statistic          N   Mean  St. Dev.  Min   Max
## --------------------------------------------------
## rank             158 79.494  45.754    1    158
## score            158 5.376   1.145   2.839 7.587
## standard_error   158 0.048   0.017   0.018 0.137
```

```
## GDP               158 0.846   0.403   0.000 1.690
## family            158 0.991   0.272   0.000 1.402
## health            158 0.630   0.247   0.000 1.025
## freedom           158 0.429   0.151   0.000 0.670
## gov_trust         158 0.143   0.120   0.000 0.552
## generosity        158 0.237   0.127   0.000 0.796
## dystopia_residual 158 2.099   0.554   0.329 3.602
## ---------------------------------------------------
```

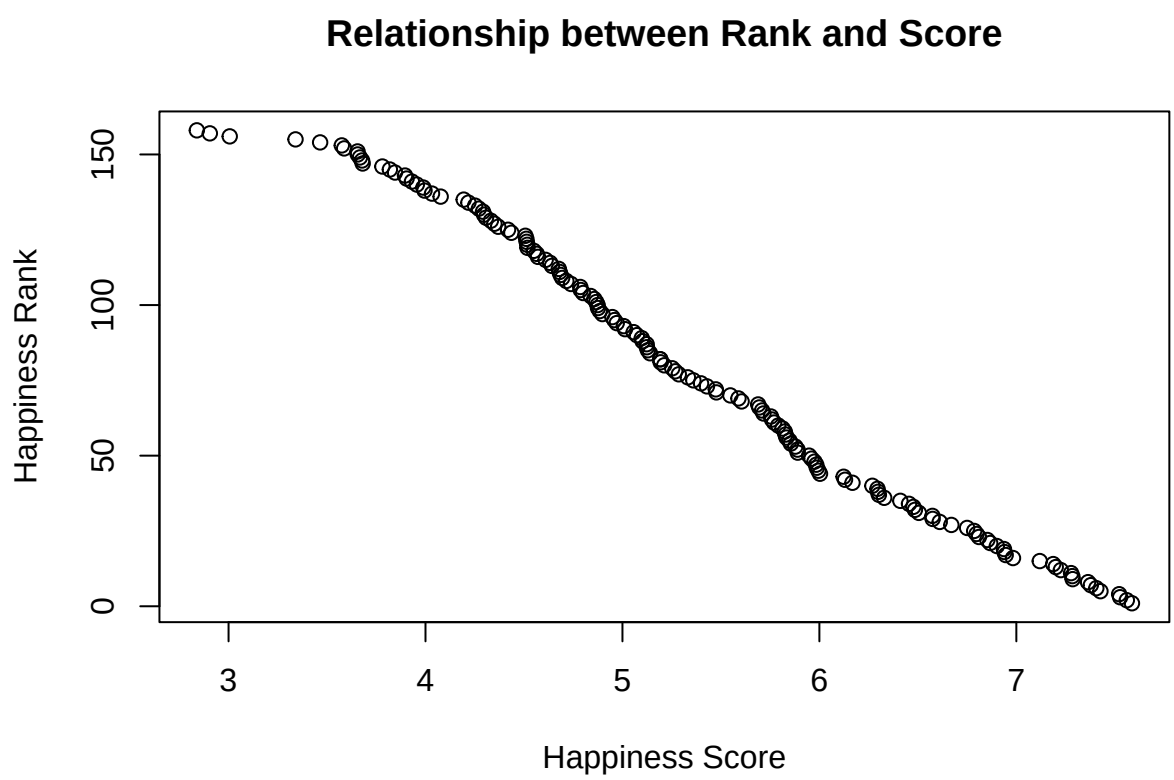
What we appear to be observing in this dataset are the factors contributing to happiness among different countries. There are 158 observations, each clearly being ranked and scored based on happiness levels. There are measures related to the government, such as gross domestic product (with an average value of .8461 and a median of .9102) and trust in the government (with a mean of .14342 and a median of .10722). Additionally, there are measures such as family (with a mean of .991 and a median of 1.0295) and health (with a mean of .6303 and a median of .6967), which seem to be more related to individual values.

The variable measuring happiness, score, has a fairly large range with a mean of 5.376 and a standard deviation of 1.145, suggesting there is quite some variation in the happiness levels of different countries.

gov_trust has a fairly low mean of 0.143 and the maximum value is only 0.552, meaning across the board, even the happiness countries do not have much trust in their government. family, health, and freedom however have fairly high means in relation to their maximum and minimum values.

Now produce a scatter plot for the 'rank' column (y) and that of the 'score' (x).

```r
plot(new_data$score, new_data$rank,
     xlab = "Happiness Score", ylab = "Happiness Rank",
     main = "Relationship between Rank and Score")
```



We can clearly observe from this graph that there is an inverse relationship between the happiness rank and the happiness score (as the score increases, the rank overall decreases). From this, it appears that these two variables give us the same information, simply represented differently.

For this scatter plot, let's adjust the annotations, tick sizes, title etc, to improve the readability.

```
par(mar = c(5, 5, 4, 2)) # Adjusts parameters of our plot
plot(new_data$score, new_data$rank,
     xlab = "Happiness Score", ylab = "Happiness Rank",
     main = "Relationship between Rank and Score",
     col = "red", pch = 16, cex = 1.2,
     cex.lab = 1.3, cex.main = 1.5, cex.axis = 1.2,
     xaxt = "n", yaxt = "n")

# Here, I specified the color, plot character, and size
# of axis/tick labels. Additionally, I removed the
# automatic labeling of the axes tick marks, so that I
# could create them myself, as seen below.

axis(2, at = seq(0, 150, by = 25))
axis(1, at = seq(2, 8, by = .5))

m1 <- lm(rank ~ score, data = new_data)
abline(m1, lwd = 2)
```

```r
summary(m1)
```

```
##
## Call:
## lm(formula = rank ~ score, data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.0608  -3.9687   0.0754   4.5849   9.5936
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 292.6110     2.2049  132.71   <2e-16 ***
## score       -39.6443     0.4012  -98.81   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.756 on 156 degrees of freedom
## Multiple R-squared:  0.9843, Adjusted R-squared:  0.9842
## F-statistic:  9763 on 1 and 156 DF,  p-value: < 2.2e-16
```

On the scatter plot, let's add a fitted line through the points and discuss the quality of that fit (where the independent is the 'happiness score' and the dependent is the 'happiness rank')

The summary of our model informs us this linear model represents our data quite well, yielding an R-squared of approximately 98% (telling us that the Happiness Score explains 98% of the variation in our dependent variable). The independent variable is extremely statistically significant as well. Though the line is not a perfect representation of our data, we can clearly see that it highlights the trend of these two variables with each other (again, an inverse relationship).

Now, we should produce a linear model with the independent as the 'happiness score' and the dependent the 'Health' variable.

```r
lhs_vs_health <- lm(health ~ score, data = new_data)
summary(lhs_vs_health)
```

```
##
## Call:
## lm(formula = health ~ score, data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49450 -0.09065  0.00356  0.10157  0.46199
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.20982    0.06547  -3.205  0.00164 **
## score        0.15627    0.01191  13.117  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1709 on 156 degrees of freedom
## Multiple R-squared:  0.5245, Adjusted R-squared:  0.5214
## F-statistic: 172.1 on 1 and 156 DF,  p-value: < 2.2e-16
```

Using "Health" as the dependent variable, the happiness score is a statistically significant explanatory variable at the 95% significance level. At the same time, we can see that it explains a little over half of the variance in our dependent variable (52.45%).

Now, let's try fitting a degree 5 polynomial to the data and plot the model fit.

```
phs_vs_health <- lm(health ~ poly(score, degree = 5),
                    data = new_data)
summary(phs_vs_health)
```

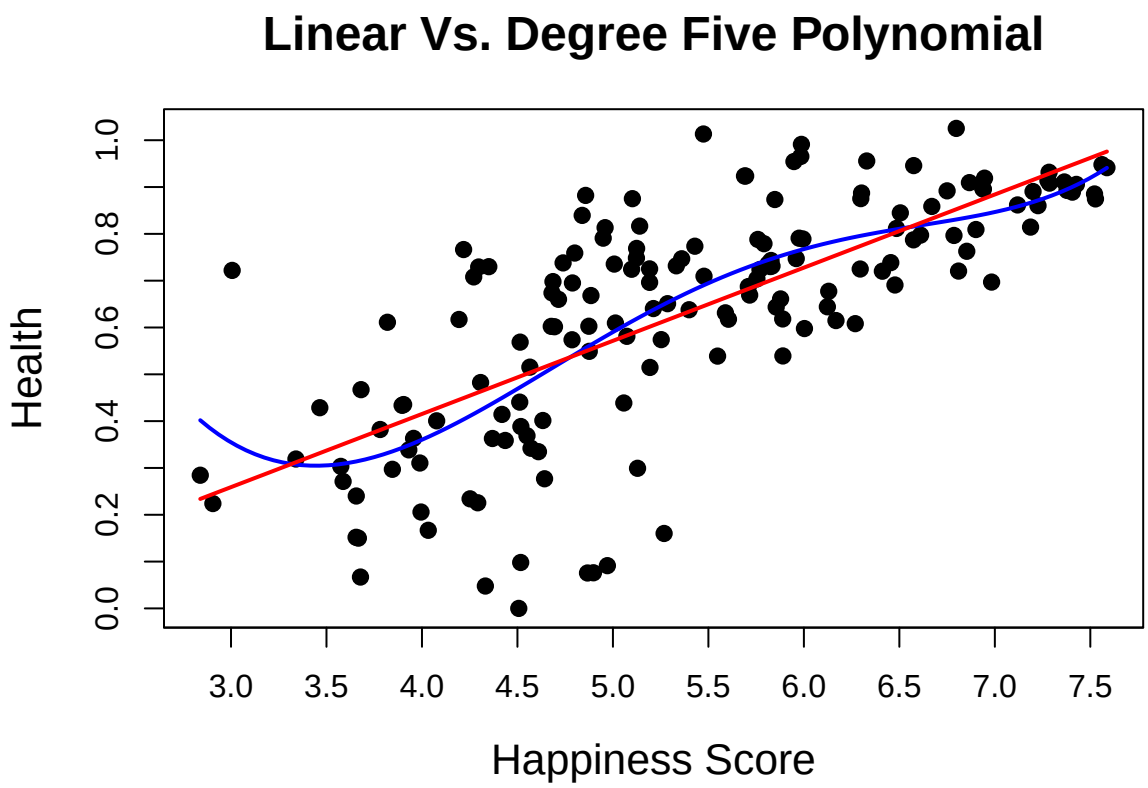
```
##
## Call:
## lm(formula = health ~ poly(score, degree = 5), data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49212 -0.08469  0.00097  0.09121  0.36871
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               0.63026    0.01339  47.079   <2e-16 ***
## poly(score, degree = 5)1  2.24203    0.16828  13.324   <2e-16 ***
## poly(score, degree = 5)2 -0.19591    0.16828  -1.164   0.2462
## poly(score, degree = 5)3 -0.35215    0.16828  -2.093   0.0380 *
## poly(score, degree = 5)4  0.30097    0.16828   1.789   0.0757 .
## poly(score, degree = 5)5  0.02500    0.16828   0.149   0.8821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1683 on 152 degrees of freedom
## Multiple R-squared:  0.5509, Adjusted R-squared:  0.5362
## F-statistic: 37.29 on 5 and 152 DF,  p-value: < 2.2e-16
```

Our intercept and "score" variable raised to the first power are both extremely significant at a significance/alpha level of .001 (99.9% confidence level). score raised to the third power is significant at a significance level of .05 (95% confidence level), while score raised to the fourth power is significant at a level of .1 (90% confidence level).

The significance level of these variables tells us that the polynomial does somewhat better represent our data, but the linear component is the most relevant part.

```
par(mar = c(5, 5, 4, 2))
plot(new_data$score, new_data$health,
     xlab = "Happiness Score",
     ylab = "Health",
     main = "Linear Vs. Degree Five Polynomial",
     col = "black", pch = 16, cex = 1.2,
     cex.lab = 1.3, cex.main = 1.5, cex.axis = 1.2,
     xaxt = "n", yaxt = "n")
seq <- seq(min(new_data$score), max(new_data$score), length.out = 100)
predict<- predict(phs_vs_health, newdata = data.frame(score = seq))
predict2 <- predict(lhs_vs_health, newdata = data.frame(score = seq))
lines(seq, predict, col = "blue", lwd = 2)
lines(seq, predict2, col = "red", lwd = 2)
```

```r
axis(2, at = seq(0, 1, by = .1))
axis(1, at = seq(2, 8, by = .5))
```



Exploring both summaries, as well as the visuals, we see that the polynomial model better represents our data, but only slightly. The R-squared increases from 0.5245 to 0.5509, telling us that the polynomial model explains a bit more of our data than the linear one.

Moreover, looking at the plot, it's apparent that the data is not exactly linear. The polynomial better follows the range where the data dips downward and the range where the data rises.

The F-statistic is very large with an extremely low p-value, telling us that this model as a whole explains more of the variance than a model with no predictor variables. Collectively, our polynomial model has some significance, even if a few of the polynomial terms are not significant.

Now let's try using some regression using multiple variables. Let's start with a linear model using all the variables specified and calculating the corresponding root mean squared error.

```r
multi_lm <- lm(score ~ GDP + family + health + freedom
               + gov_trust + generosity + dystopia_residual,
               data = new_data)
summary(multi_lm)
```

```
##
## Call:
## lm(formula = score ~ GDP + family + health + freedom + gov_trust +
##     generosity + dystopia_residual, data = new_data)
##
```

```
## Residuals:
##        Min         1Q     Median         3Q        Max
## -5.701e-04 -2.224e-04 -2.580e-06  2.471e-04  5.054e-04
##
## Coefficients:
##                     Estimate Std. Error   t value Pr(>|t|)
## (Intercept)       6.405e-05  1.245e-04     0.514    0.608
## GDP               1.000e+00  1.129e-04  8855.750   <2e-16 ***
## family            1.000e+00  1.153e-04  8675.863   <2e-16 ***
## health            9.999e-01  1.619e-04  6175.103   <2e-16 ***
## freedom           9.997e-01  1.976e-04  5059.468   <2e-16 ***
## gov_trust         9.999e-01  2.236e-04  4470.866   <2e-16 ***
## generosity        1.000e+00  2.018e-04  4956.272   <2e-16 ***
## dystopia_residual 1.000e+00  4.166e-05 24003.904   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0002821 on 150 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 3.695e+08 on 7 and 150 DF,  p-value: < 2.2e-16
```

```r
multi_lm_rmse <- sqrt(mean(multi_lm$residuals^2))
multi_lm_rmse
```

```
## [1] 0.0002748618
```

The RMSE for this model is incredibly small, only 0.0002748618. Thus, this linear model with multiple variables does a good job fitting the data and making accurate predictions. Every predictor variable is statistically significant at a 99.9% confidence level. Moreover, the R-squared is equal to 1, and the F-statistic is drastically high, implying that there is no way for the improvement of model to go but down.

Now applying LASSO:

```r
x <- model.matrix(score ~ GDP + family + health + freedom
                  + gov_trust + generosity + dystopia_residual,
                  data = new_data)[, -1]
y <- new_data$score

cv_lasso <- cv.glmnet(x, y, alpha = 1)
penalty_opt <- cv_lasso$lambda.min
print(penalty_opt)
```

```
## [1] 0.01790988
```

```r
opt_lasso <-glmnet(x, y, alpha = 1, lambda = penalty_opt )
coef(opt_lasso, s= cv_lasso$lambda.min)
```

```
## 8 x 1 sparse Matrix of class "dgCMatrix"
##                           s1
## (Intercept)       0.1574639
## GDP               0.9818996
## family            0.9837081
```

```
## health            0.9783605
## freedom           1.0060771
## gov_trust         0.9165783
## generosity        0.8690188
## dystopia_residual 0.9657260
```

Relative to each other, the coefficients of all our variables are very close. This adds more confirmation that each of these predictors adds value to our model. Under this notion, it would not make sense to eliminate any one of them.

```
lasso_predict <- predict(opt_lasso, newx = x, type = "response",
                         s = opt_lasso)
lasso_rmse <- sqrt(mean((y - lasso_predict)^2))
lasso_rmse
```

```
## [1] 0.03317726
```

Using the optimal lambda value, the RMSE comes out to 0.03317726, which is greater than the RMSE of the previous model. While applying LASSO did adjust some of the coefficient values (though not drastically), none of them were shrunk to 0, so they were all still included in this LASSO model.

Next, let's produce a support vector machine model with the same variables:

```
x <- new_data[, c("GDP", "family", "health", "freedom",
                  "gov_trust", "generosity", "dystopia_residual")]
y <- new_data$score

svm <- svm(score ~ GDP + family + health + freedom
           + gov_trust + generosity + dystopia_residual,
           data = new_data)
summary(svm)
```

```
##
## Call:
## svm(formula = score ~ GDP + family + health + freedom + gov_trust +
##     generosity + dystopia_residual, data = new_data)
##
##
## Parameters:
##    SVM-Type:  eps-regression
##  SVM-Kernel:  radial
##        cost:  1
##       gamma:  0.1428571
##     epsilon:  0.1
##
##
## Number of Support Vectors:  53
```

```
svm_predict <- predict(svm, new_data)
svm_rmse <- RMSE(svm_predict, y)
svm_rmse
```

```
## [1] 0.1096093
```

The RMSE for our support vector machine model is 0.1096093, an increase from both the models we previously observed. Based on this measure, the SVM model does worse at prediction than both the linear model and the linear w/ lasso model.

Finally, let's look at the Pearson correlation, Spearman rank correlation, and Mutual Information.

```
pearson_cor <- cor(new_data[, c("score", "GDP", "family", "health",
                                "freedom", "gov_trust", "generosity",
                                "dystopia_residual")], method = "pearson")
pearson_cor
```

```
##                       score         GDP     family     health    freedom
## score             1.0000000  0.78096553 0.74060520 0.72419960 0.56821090
## GDP               0.7809655  1.00000000 0.64529939 0.81647800 0.37029971
## family            0.7406052  0.64529939 1.00000000 0.53110399 0.44151821
## health            0.7241996  0.81647800 0.53110399 1.00000000 0.36047653
## freedom           0.5682109  0.37029971 0.44151821 0.36047653 1.00000000
## gov_trust         0.3951986  0.30788520 0.20560511 0.24833468 0.49352352
## generosity        0.1803185 -0.01046544 0.08751324 0.10833522 0.37391613
## dystopia_residual 0.5304735  0.04005876 0.14811704 0.01897932 0.06278344
##                     gov_trust  generosity dystopia_residual
## score              0.39519858  0.18031853        0.53047352
## GDP                0.30788520 -0.01046544        0.04005876
## family             0.20560511  0.08751324        0.14811704
## health             0.24833468  0.10833522        0.01897932
## freedom            0.49352352  0.37391613        0.06278344
## gov_trust          1.00000000  0.27612270       -0.03310457
## generosity         0.27612270  1.00000000       -0.10130111
## dystopia_residual -0.03310457 -0.10130111        1.00000000
```

The pearson coefficient is a linear measure of correlation, ranging between -1 and +1.

```
pcor_df <- as.data.frame(as.table(pearson_cor))
neg_pcor_df <- pcor_df[pcor_df$Freq < 0, ]
neg_pcor_df
```

```
##                 Var1              Var2        Freq
## 15        generosity               GDP -0.01046544
## 48 dystopia_residual         gov_trust -0.03310457
## 50               GDP        generosity -0.01046544
## 56 dystopia_residual        generosity -0.10130111
## 62         gov_trust dystopia_residual -0.03310457
## 63        generosity dystopia_residual -0.10130111
```

The majority of our variables have a positive linear correlation, with only three negatively correlated relationships (those between GDP and generosity, dystopia_residual and gov_trust, and generosity and dystopia_residual).

```
pearson_cor[lower.tri(pearson_cor, diag = TRUE)] <- NA

max_pcor <- max(pearson_cor, na.rm = TRUE)
max_pcor
```

```
## [1] 0.816478
```

```r
which(pearson_cor == max_pcor, arr.ind = TRUE)
```

```
##     row col
## GDP   2   4
```

The variables with maximum positive correlation are GDP and health, with a correlation value of 0.816478.

```r
min_pcor <- min(pearson_cor, na.rm = TRUE)
min_pcor
```

```
## [1] -0.1013011
```

```r
which(pearson_cor == min_pcor, arr.ind = TRUE)
```

```
##            row col
## generosity   7   8
```

The variables with maximum negative correlation are generosity and dystopia_residual, with a correlation value of -0.10130111.

Similarly, Spearman Rank correlation also returns a value between -1 and +1. It is a non-parametric measure of rank correlation, observing how well a monotonic function describes the relationship between two variables.

```r
spearman_cor <- cor(new_data[, c("score", "GDP", "family", "health",
                                 "freedom", "gov_trust", "generosity",
                                 "dystopia_residual")], method = "spearman")
spearman_cor
```

```
##                      score        GDP    family     health   freedom
## score            1.0000000 0.79570280 0.7700380 0.75696044 0.5763522
## GDP              0.7957028 1.00000000 0.6810419 0.84709637 0.4154214
## family           0.7700380 0.68104187 1.0000000 0.58543093 0.5281391
## health           0.7569604 0.84709637 0.5854309 1.00000000 0.3998001
## freedom          0.5763522 0.41542141 0.5281391 0.39980011 1.0000000
## gov_trust        0.3023969 0.22767485 0.1845994 0.16884711 0.4637454
## generosity       0.1724630 0.01248176 0.1463995 0.11025643 0.4184578
## dystopia_residual 0.5324872 0.07742953 0.1905336 0.07833321 0.0954410
##                    gov_trust  generosity dystopia_residual
## score            0.30239687 0.172462971       0.532487220
## GDP              0.22767485 0.012481764       0.077429532
## family           0.18459942 0.146399456       0.190533635
## health           0.16884711 0.110256428       0.078333208
## freedom          0.46374543 0.418457799       0.095441000
## gov_trust        1.00000000 0.213356943       0.054814838
## generosity       0.21335694 1.000000000       0.008091469
## dystopia_residual 0.05481484 0.008091469       1.000000000
```

One might note that there are no negative correlations in this matrix, telling us that all the variables in our dataset have increasing relationships with one another (when one variable goes up, the other variable goes up as well)

```r
spearman_cor[lower.tri(spearman_cor, diag = TRUE)] <- NA

max_scor <- max(spearman_cor, na.rm = TRUE)
max_scor
```

```
## [1] 0.8470964
```

```r
which(spearman_cor == max_scor, arr.ind = TRUE)
```

```
##     row col
## GDP   2   4
```

The variables with maximum positive rank correlation are GDP and health, with a correlation value of 0.8470964. These are the same variables with maximum linear correlation.

```r
min_scor <- min(spearman_cor, na.rm = TRUE)
min_scor
```

```
## [1] 0.008091469
```

```r
which(spearman_cor == min_scor, arr.ind = TRUE)
```

```
##            row col
## generosity   7   8
```

The variables with the least rank correlation are generosity and dystopia_residual. Interestingly, these variables had the greatest negative linear correlation previously; however, now it is expressed that there is no negative relationship between the two. Essentially, there is barely any monotonic relationship between these two predictor variables, according to the Spearman Rank results.

Finally, mutual information is a measure of how much one variable tells us about another variable. Any values close to zero indicate a very small relationship between variables (and if the mutual information = 0, then the variables are entirely independent). The larger the mutual information, the more certainty we might acquire about one random variable given information about another. A benefit of Mutual Information is that it examines both linear and nonlinear dependencies.

```r
discretized<- discretize(new_data[, c("score", "GDP", "family",
                                      "health", "freedom", "gov_trust",
                                      "generosity", "dystopia_residual")])

mi_matrix <- mutinformation(discretized)
mi_matrix
```

```
##                       score        GDP     family     health    freedom
## score             1.60852583 0.43962692 0.41881987 0.38481605 0.26194799
## GDP               0.43962692 1.60852583 0.32361607 0.60550065 0.16925611
## family            0.41881987 0.32361607 1.60852583 0.27521070 0.19671736
## health            0.38481605 0.60550065 0.27521070 1.60852583 0.18063161
## freedom           0.26194799 0.16925611 0.19671736 0.18063161 1.60852583
## gov_trust         0.17129971 0.27066173 0.10132865 0.14248900 0.20439840
```

```
## generosity        0.08317026 0.16988832 0.09193224 0.13753698 0.13051732
## dystopia_residual 0.22899274 0.06322446 0.12251167 0.02725755 0.06805172
##                    gov_trust generosity dystopia_residual
## score             0.17129971 0.08317026        0.22899274
## GDP               0.27066173 0.16988832        0.06322446
## family            0.10132865 0.09193224        0.12251167
## health            0.14248900 0.13753698        0.02725755
## freedom           0.20439840 0.13051732        0.06805172
## gov_trust         1.60852583 0.05848975        0.09340989
## generosity        0.05848975 1.60852583        0.03109660
## dystopia_residual 0.09340989 0.03109660        1.60852583
```

```
mi_matrix[lower.tri(mi_matrix, diag = TRUE)] <- NA

max_mi <- max(mi_matrix, na.rm = TRUE)
which(mi_matrix == max_mi, arr.ind = TRUE)
```

```
##     row col
## GDP   2   4
```

Maximum mutual information is found between GDP and health. Considering they had the strongest correlation for both Pearson and Spearman Rank, this further indicates a strong association between these two variables.

```
min_mi <- min(mi_matrix, na.rm = TRUE)
which(mi_matrix == min_mi, arr.ind = TRUE)
```

```
##        row col
## health   4   8
```

Minimum mutual information is found between heatlh and dystopia_residual. We might also rank all of these values to determine where each pair of variables lies in terms of who has the the most/least mutual information:

```
mi_rank <- as.data.frame(as.table(mi_matrix))
mi_rank <- mi_rank[mi_rank$Var1 != mi_rank$Var2 & !is.na(mi_rank$Freq), ]
mi_rank <- mi_rank[!duplicated(t(apply(mi_rank, 1, sort))), ]
mi_rank <- mi_rank[order(-mi_rank$Freq), ]
mi_rank
```

```
##          Var1              Var2       Freq
## 26        GDP            health 0.60550065
## 9       score               GDP 0.43962692
## 17      score            family 0.41881987
## 25      score            health 0.38481605
## 18        GDP            family 0.32361607
## 27     family            health 0.27521070
## 42        GDP         gov_trust 0.27066173
## 33      score           freedom 0.26194799
## 57      score dystopia_residual 0.22899274
## 45    freedom         gov_trust 0.20439840
```

```
## 35     family           freedom 0.19671736
## 36     health           freedom 0.18063161
## 41      score         gov_trust 0.17129971
## 50        GDP        generosity 0.16988832
## 34        GDP           freedom 0.16925611
## 44     health         gov_trust 0.14248900
## 52     health        generosity 0.13753698
## 53    freedom        generosity 0.13051732
## 59     family dystopia_residual 0.12251167
## 43     family         gov_trust 0.10132865
## 62  gov_trust dystopia_residual 0.09340989
## 51     family        generosity 0.09193224
## 49      score        generosity 0.08317026
## 61    freedom dystopia_residual 0.06805172
## 58        GDP dystopia_residual 0.06322446
## 54  gov_trust        generosity 0.05848975
## 63 generosity dystopia_residual 0.03109660
## 60     health dystopia_residual 0.02725755
```

From this, we can determine that two of the variables that tend to be more independent from each of the others are dystopia_residual and generosity, indicating that they do not contribute as much to any of the other variables.